

ChE 210 Homework Set #5 Due on Tuesday, 18 February 2020

(1) Rework Murphy 2.67. Use the same notation that I did in class labeling the species as A , B , C , D , E , and I . However, the Oyster Shell data has been changed so that the composition is 2.1 wt% A , 55 wt% B and 42.9 wt% I .

- (a) Set up species balances for each of the six species listed above and apply the appropriate constraint equation. Treat the weight fraction of D in stream 6 as a design variable with a constrained range. Express this 7 X 7 system of equations in matrix form. Solve the system using any method you deem best.
- (b) Could the equations be solved by successive substitution as I did in lecture? Make a table of variables and equations indicating in which equations each variable appears. Discuss your answer.

(2) Murphy 3.21

(3) Murphy 3.24

(4) Murphy 3.29

(5) Murphy 3.41